

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
First Semester 2023-2024

Comprehensive Examination
(EC-3 Regular)

Course No. : DSECLZG522
Course Title : Big Data Systems
Nature of Exam : Open Book
Weightage : 40%
Duration : 2.5 Hours
Date of Exam : 06-04-2024 (AN)

No. of Pages	= 2
No. of Questions	= 8

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made, if any, should be stated clearly at the beginning of your answer.

Q.1. What are the factors contributing to MTBF? Define Availability. When do you say that a system is highly available? A middleware product has memory leaks and it core dumps every 10 hours on an average. Upon failure, it is automatically restarted and will be back into operation within 3 minutes. Find out the availability of the middleware product?

[4 Marks]

Q.2. What is the purpose of database sharding? What is the difference between sharding and replication?

[4 Marks]

Q.3. An enterprise company need to address the following challenges with Big Data Analytics:

- (a) Struggling with a Java based ELT tool to manage and extract value from the growing volume and variety of data and need to unify information across federated sources. Each disparate MySQL server records around 2 million updates, a total of 50 million updates daily.
 - (b) Unable to relate raw data collected from system logs, sensors, or click streams with customer and line-of-business data managed in enterprise systems.
 - (c) Each end-to-end ELT process consumes 14 hours. This process should be completed within an hour.
 - (d) Some data warehousing queries on structured data are available only in SQL and there is no expertise available for transforming these SQL queries into Hadoop eco system tools.
- Recommend suitable solutions to address the above challenges by making use of Hadoop Ecosystem products/tools. Give justifications for your recommendations.

[6 Marks]

Q.4. You are given access to 10 Terabyte of data with details of daily sales transactions belonging to an e-commerce company. The transactional data files are stored in a folder on HDFS with a structure as shown below:

Transaction No. (Unique)	Invoice	Stock Code	Description	Quantity	Invoice Date	Price(\$)	Customer ID	Country
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It is not required to modify the data. You are not allowed to copy or move the data into any other storage location. Select a suitable analytics tool from the Hadoop ecosystem family to run queries on this data. Give justifications for your selection. Using the tool selected by you, write a query to find the total sales revenue received in \$ from each country.

[6 Marks]

Q.5. In a 3 node Cassandra cluster, a Keyspace is created with a replication factor of 3.

Answer the following questions, in each of the scenarios mentioned below:

(a). Consistency of reads. (b). How many nodes can be lost without data loss.

(c). What percentage of data is held in each of the nodes.

Scenario 1 : Write Level - Quorum, Read Level - One

Scenario 2: Write Level - All, Read Level - One

Scenario 3: Write Level - One, Read Level - All

Scenario 4: Write Level - Two, Read Level - One

[4 Marks]

Q.6. Examine the Spark scala code given below:

```
val lines = new Array[String](3)
lines(0) = "zero one reject"
lines(1) = "one two reject"
lines(2) = "two two zero"
val stringRDD=sc.parallelize(lines,2)
println("Number of Partitions: "+stringRDD.getNumPartitions)
val wordsRDD = stringRDD.flatMap(x => x.split(" "))
val filteredRDD = wordsRDD.filter(x => x.startsWith("o") || x.startsWith("t"))
val wcRDD = filteredRDD.map(word => (word, 1)).reduceByKey(_ + _)
wcRDD.collect().foreach(println)
```

(a) What is the output of this code ?

(b) Describe the RDD lineage in the above code

[4 Marks]

Q.7. What are the capabilities of SparkSQL ?

The "bank.csv" file has customer data with the following schema:

AcctNo	age	job	Education	balance	loan	name	contact
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Read data from this file into dataframe and create a table view. Write Spark SQL queries to:

(a). Find the total number of customers within each job category.

(b). Find age, education, balance of customers with admin job.

[6 Marks]

Q.8. Why is windowing needed in Stream computing? Data from a smart device is transmitted over TCP/IP connection to a Spark streaming analytics system. Data from the device arrives at the rate of 1 packet per second. The RDD batching interval is set to 5 seconds. The stream analytics system uses a stateful algorithm with a time window of 60 seconds. It is required to generate an output from the streaming system every 30 seconds. What type of windowing is recommended by you? Give appropriate configuration parameters for the Window suggested by you. What will be the number of RDDs in the Dstreams generated in one window duration.

[6 Marks]
